

WinDbg cheatsheet

Command	Notes
?, .help, .hh	help
.symopt+0x100	disable auto symbol loading
.symopt-0x100	re-enable auto symbol loading
pt	execute to return
ph [n]	execute up to N branch instructions
bp [addr] ["script"]	break
bu symbol	set unresolved breakpoint
bm pattern	pattern match
ba [e/r/w] size address	memory access breakpoint
bl	list breakpoints
be 1	enable
bd 3	disable
lm	list modules
.sympath+ path	add to symbol path
.srcpath+ path	add to source path
.reload /f	reload symbols
ld m	load symbols for modules
x module!sympat	list symbols matching patterns
ln address	find nearest symbol to address
r reg=value	assign value to register
dW addr L#	display memory; W is many things
dx (libtypes!My::CustomString*)&tmpNam2	display with natvis, cast to type
dv /t	list locals and their types
dx a,20	display 20 array elements
dx s,s8	display as utf-8 string
dx utf16s,su	display as utf-16 string
dx i,x	display i in hex
dx i,d	display i in decimal
dv /f libmod!NS::Func	display parameters and locals for any f
e{b d D f p q w} Address [Values]	edit memory commands
e{a u z zu} Address "String"	c.f. "Accessing Memory by Virtual Address"
e Address [Values]	
u @rbx L8	unassemble RBX register for 8 bytes
s -d addr1 addr2 41414141	search memory
s -a addr1 addr2 "foobar"	search memory
dt ntdll!_TEB @\$teb,dt my.exe!potato_s @\$rax	structured print
? expr	evaluates expression
?? expr	evaluates C++ expression
0x, 0n, 0y	hex, decimal, binary prefix
kp	stack trace with all function parameters

Command	Notes
a addr	start assembler and write starting at addr
sxe eh	always break on C++ exceptions
sxd eh	second-chance break
!analyze, !analyze -hang	magic!
.frame	set local context
!tt.index	rebuild timetravel index from .run file
p-, t-, g-	step (F10), trace (F11), go (F5) but backwards!
!positions	thread positions at given time
!tt 200:0	time travel to position 200
dx -r1 @\$curprocess.TTD.Events	list all events in recording
CTRL + BREAK	interrupt debugger
as qq .shell -i- worddb_query f \${\$arg1}	create an alias with parameters

Tips:

- CTRL+BREAK tells it to stop loading, but it takes a while
- Use x libthing!*Symbo* to list symbols to check for precise things when setting breakpoints
- It picks up srcpaths's if, when broken into, you go to Source → Open Source File and paste in the value returned by qudu qwhere -u fname
- You can
 - bu `libthing!C:\src\thing\thing.c:657`
- File → Settings → Debugging → Startup:

```
.symopt+0x100
as qq .shell -i- worddb_query f ${$arg1}
.nvload "C:\Users\user\CustomVisualizers.natvis"
```

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